hw03_version2

March 17, 2025

1 Homework 3: Bayesian Models and Monte Carlo

BEE 4850/5850, Spring 2025

Name:

ID:

> **Due Date**
>
> Friday, 3/14/25, 9:00pm

1.1 Overview

1.1.1 Instructions

The goal of this homework assignment is to practice developing and working with probability models for data.

- Problem 1 asks you to quantify uncertainties for a simple model using Bayesian statistics and rejection sampling.
- Problem 2 asks you to use Monte Carlo simulation to calculate annual expected flooding damages.
- Problem 3 (only required for students in BEE 5850) asks you to conduct cost-benefit analyses of housing elevation levels under future flooding uncertainty using Monte Carlo simulations.

1.2 QUICK NOTE ABOUT REPRODUCIBILITY/EXACT VALUES IN RESULTS BELOW:

I set the random seed at the top of the document, but realized that wasn't resuling in the same numbers each time I reset the kernel and ran the entire code. So I tried setting the random seed multiple times throughout the document (essenitally in a code block any time random values are drawn), and that still didn't help. I think a possible fix is setting the random seed via numpy and feeding it into each randomized function, but I chatted with Vivek and he said it was ok if I didn't fix this right now. So - some of the values I reference in my commentary (that is in markdown code blocks) might be slightly different than the values output in print statements from the code, but in all the cases I've seen, these are very minor differences.

1.2.1 Load Environment

USED CHATGPT (OpenAI, 2025): converted Julia to Python for start of script

[1]:
```
import random  # random number generation and seed-setting
import pandas as pd  # tabular data structure
import csv  # reads/writes .csv files (though pandas is usually used for this)
import numpy as np  # numerical computing
import scipy.stats as stats  # interface to work with probability distributions
import matplotlib.pyplot as plt  # plotting library
import seaborn as sns  # additional statistical plotting tools
import statistics  # statistical quantities like mean, median, etc.
from scipy.optimize import minimize  # optimization tools
random.seed(44)
```

1.3 Problems

1.3.1 Scoring

- Problem 1 is worth 13 points.
- Problem 2 is worth 7 points.
- Problem 3 is worth 5 points.

1.3.2 Problem 1

Consider the class-cheating testing procedure from Homework 1. As a reminder, the procedure is as follows:

> Each student flips a fair coin, with the results hidden from the interviewer. The student answers honestly if the coin comes up heads. Otherwise, if the coin comes up tails, the student flips the coin again, and answers "I did cheat" if heads, and "I did not cheat", if tails.

Suppose that, after conducting the procedure on your class of 100 students, you get 40 "Yes, I cheated" responses. You are skeptical that your class is broadly engaged in cheating and have a prior belief that the probability of cheating θ is unlikely to be high. We'll express this prior using a beta distribution, $(3, 40), which is shown below. You would like to understand the range of cheating propensity which is consistent with your prior beliefs and the outcomes from the detection procedure.

In this problem:

- A) Construct a Bayesian probability model (based on the data-generating process from Homework 1 and our prior on the cheating probability) for the "Yes, I cheated" counts. You can compute the likelihood based on the combination of the fair coin flip and the "true" cheating probability θ.
- B) Plot the posterior $p(\theta|y)$ using a grid approximation (looping over values of θ).
- C) Use rejection sampling to draw 1,000 samples from the posterior and plot the distribution of samples. How well are you capturing the posterior you obtained from the grid approximation? Would you prefer to have more samples?
- D) Report the expected value of θ and its 90% credible interval. What conclusions can you draw about cheating probability and whether your belief in your class was warranted? How much uncertainty exists as a result of this privacy-protecting procedure?

- E) Repeat the analysis with a less constrained prior of your choosing. How does that change your inferences and conclusions? What does this tell you about the effectiveness of the procedure?

USED CHATGPT (OpenAI, 2025): I had used chatgpt to help create the convergence plot I made in problem #3 (where it wasn't required; explanation below at problem 3). And after creating it, I realized it would also be useful to visualize a similar thing for quesiton 1 and 2, so I adapted the code accordingly. I also used chatgpt to help create secondary y axes for my likelihood/prior/posterior plot, but the y axis was the only thing modified by chatgpt.

A) Construct a Bayesian probability model (based on the data-generating process from Homework 1 and our prior on the cheating probability) for the "Yes, I cheated" counts. You can compute the likelihood based on the combination of the fair coin flip and the "true" cheating probability θ.

Recall from hw 1: Each student flips a fair coin, with the results hidden from the > interviewer. The student answers honestly if the coin comes up heads. > Otherwise, if the coin comes up tails, the student flips the coin > again, and answers "I did cheat" if heads, and "I did not cheat", if > tails.

Outcomes: H –> answer honest –> the number here that say "I did cheat" would be p_fair * p_cheat = $1/2p\_cheat$ *T –> flip agian, and IF it comes up heads, say "i did cheat" so this would be p_fair* p_fair = 1/4 So the associated probability in one single binomial distributiont that represents this, would be 1/2*p_cheat + 1/4

Likelihood of a given theta, given that 40/100 said "yes i cheated", and we know the data generating set up (i.e., probability of saying "yes i cheated" because the coin flip TOLD you to, PLUS the probabilty of saying "yes i cheated" because the coin flip said you had to be honest, and there is a "true" probability of cheating that we can recover from this.

NOTE ON APPROACH: below is my Bayesian probability model. The likelihood is the probability of observing 40/100 "yes I cheated" given the "true probability" - i.e., based on the coin experiment as noted above, this results in p = 1/2 * 1/2 + 1/2 * theta = 0.25+0.5 * theta, modeling likelihood with a binomial distribution and this value of p. This likelihood gets multiplied by the prior, i.e., the beta distribution with parameters (3, 40), evaluated at a given theta ("true probability of cheating").

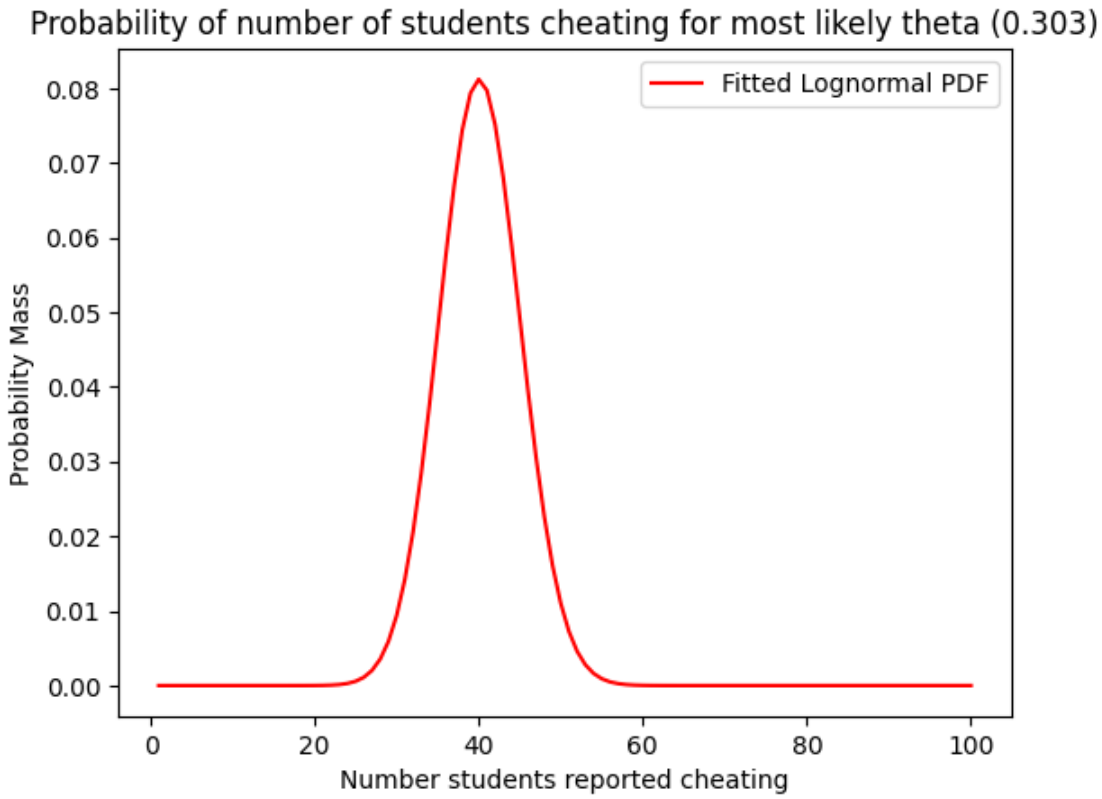
[2]:
```
# likelihood: Likelihood of a given theta, given that 40/100 said "yes i␣
 ↪cheated",
# and we know the data generating set up (i.e., probability of saying "yes i␣
 ↪cheated"
# because the coin flip TOLD you to, PLUS the probabilty of saying "yes i␣
 ↪cheated" because the coin
# flip said you had to be honest, and there is a "true" probability of cheating␣
 ↪that we can recover from this.
theta = np.linspace(0,1,100)

def find_likelihood(theta_):
    likelihood_ = stats.binom.pmf(k = 40, n = 100, p = (0.25+.5*theta_), loc=0)
    return likelihood_
```

```
likelihood = []
for i in np.arange(0, len(theta)):
    likelihood.append(find_likelihood(theta[i]))
val_max = np.max(likelihood)
idx_max = np.argmax(likelihood)
print(f'Max prob cheating {round(val_max,3)} (likelihood) at index {idx_max}')
```

```
Max prob cheating 0.081 (likelihood) at index 30
```

[3]:
```
def find_prior(theta_):
    # x, a, b, loc=0, scale=1
    prior_ = stats.beta.pdf(theta_, a = 3, b = 40, loc=0, scale=1)
    return prior_

# num_yes_cheat = np.arange(1, 101)
beta_prior = []
for i in np.arange(0, len(theta)):
    prior_ = find_prior(theta[i])
    beta_prior.append(prior_)
```

[4]:
```
# get pdf and plot
x = np.arange(1, 101)
pdf_binom = stats.binom.pmf(k=x, n=100, p=(0.25+0.5*theta[idx_max]), loc=0)
plt.plot(x, pdf_binom, 'r-', label='Fitted Lognormal PDF')
plt.xlabel('Number students reported cheating')
plt.ylabel('Probability Mass')
plt.title(f'Probability of number of students cheating for most likely theta␣
 ↪({np.round(theta[idx_max],3)})')
plt.legend()
plt.show()
```



^ The plot above just confirms that the most likely theta found, does in fact result in a binomial distribtuion that has 40/100 students saying yes as being likely.

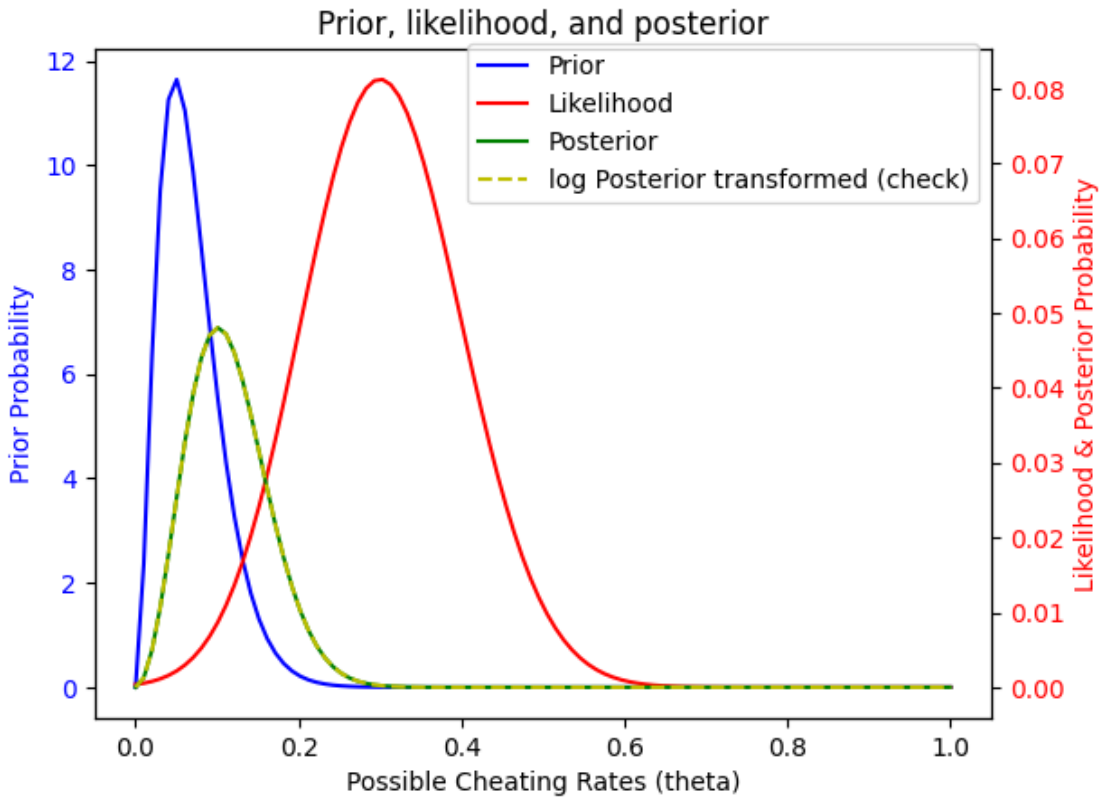
[5]:
```
df = pd.DataFrame({
    'theta': theta,
    'likelihood': likelihood,
    'beta_prior': beta_prior
})

df['posterior'] = df['likelihood']*df['beta_prior']

df['log_likelihood'] = np.log(df['likelihood'])
df['log_beta_prior'] = np.log(df['beta_prior'])

df['log_posterior'] = df['log_likelihood']+df['log_beta_prior']

df['log_posterior_transform'] = np.exp(df['log_posterior'])
```

```
/Users/mca76/Documents/Courses/EnvDataAnalysis/HW/bee5850_python_env/lib/python3
.11/site-packages/pandas/core/arraylike.py:399: RuntimeWarning: divide by zero
encountered in log
  result = getattr(ufunc, method)(*inputs, **kwargs)
```

[6]:
```
val_max = np.max(df['posterior'])
idx_max = np.argmax(df['posterior'])
print(f'Max prob cheating posterior {val_max} at index {idx_max}')

val_max = np.max(df['log_posterior'])
idx_max = np.argmax(df['log_posterior'])
print(f'Max prob cheating log-posterior {val_max} at index {idx_max},␣
 ↪exponential transform: {np.exp(val_max)}')
#print(np.exp(val_max))
```

```
Max prob cheating posterior 0.04803526241944725 at index 10
Max prob cheating log-posterior -3.0358199040470426 at index 10, exponential
transform: 0.04803526241944722
```

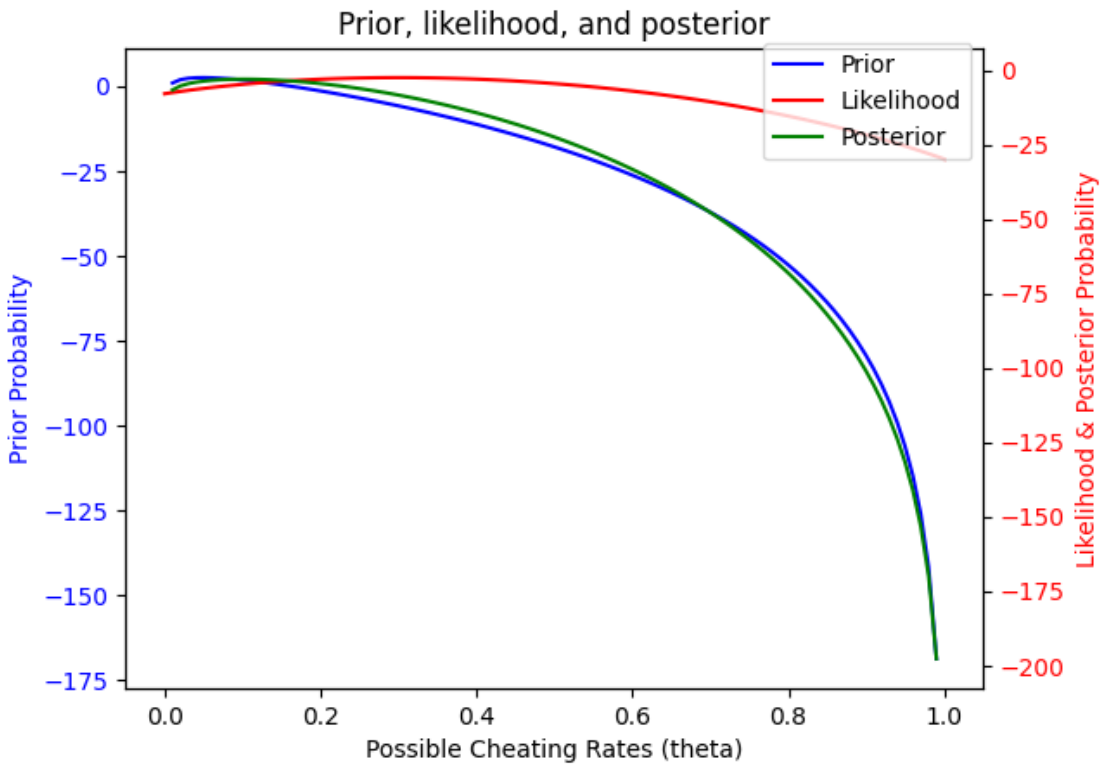
[7]:
```
fig, ax1 = plt.subplots()

# Primary y-axis (Prior)
ax1.plot(df['theta'], df['beta_prior'], 'b-', label='Prior')
ax1.set_xlabel('Possible Cheating Rates (theta)')
ax1.set_ylabel('Prior Probability', color='b')
ax1.tick_params(axis='y', labelcolor='b')

# Secondary y-axis (Likelihood & Posterior)
ax2 = ax1.twinx()
ax2.plot(df['theta'], df['likelihood'], 'r-', label='Likelihood')
ax2.plot(df['theta'], df['posterior'], 'g-', label='Posterior')
ax2.plot(df['theta'], np.exp(df['log_posterior']), 'y--', label='log Posterior␣
 ↪transformed (check)')
ax2.set_ylabel('Likelihood & Posterior Probability', color='r')
ax2.tick_params(axis='y', labelcolor='r')

# Title and Legend
plt.title('Prior, likelihood, and posterior')
fig.legend(loc='upper right', bbox_to_anchor=(.9,.9))

plt.show()
```



Note that the scales of the prior, likelihood, and posterior are a bit wonky (and not really meaningful/do not impact the analysis) - to make this visually a bit nicer I could have normalized each to their respective max densities, but I didn't bother (based on conversation with Vivek during office hours).

[8]:
```
fig, ax1 = plt.subplots()

# Primary y-axis (Prior)
ax1.plot(df['theta'], df['log_beta_prior'], 'b-', label='Prior')
ax1.set_xlabel('Possible Cheating Rates (theta)')
ax1.set_ylabel('Prior Probability', color='b')
ax1.tick_params(axis='y', labelcolor='b')

# Secondary y-axis (Likelihood & Posterior)
ax2 = ax1.twinx()
ax2.plot(df['theta'], df['log_likelihood'], 'r-', label='Likelihood')
ax2.plot(df['theta'], df['log_posterior'], 'g-', label='Posterior')
ax2.set_ylabel('Likelihood & Posterior Probability', color='r')
ax2.tick_params(axis='y', labelcolor='r')

# Title and Legend
```

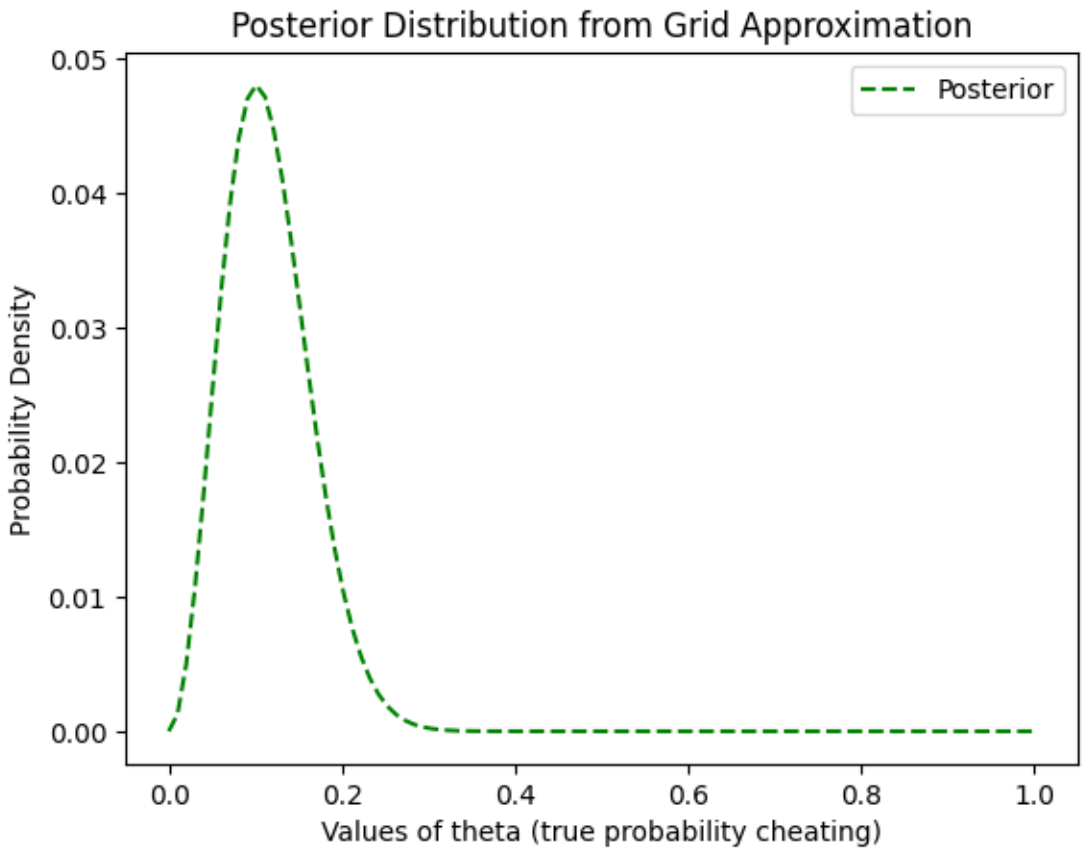
```
plt.title('Prior, likelihood, and posterior')
fig.legend(loc='upper right', bbox_to_anchor=(.9,.9))

plt.show()
```



^ The results in the two figures above and printed values before them show that you get the same answer for this problem if you 1) multiply the likelihood and prior, or 2) add the log likelihood to the log prior. (I just wanted to verify either works for this - note that for more complex problems I know I should add the log likelihood and log prior to avoid underflow issues with super small numbers resulting from two small probabilities being multiplied together).

B) Plot the posterior $p(\theta|y)$ using a grid approximation (looping over values of θ)

[9]:
```
plt.plot(df['theta'], np.exp(df['log_posterior']), 'g--', label='Posterior')
plt.xlabel('Values of theta (true probability cheating)')
plt.ylabel('Probability Density')
plt.title(f'Posterior Distribution from Grid Approximation')
plt.legend()
plt.show()
```



^ The plot above shows the posterior $p(\theta|y)$ obtained via grid approximation (looping over values of θ).

C) Use rejection sampling to draw 1,000 samples from the posterior and plot the distribution of samples. How well are you capturing the posterior you obtained from the grid approximation? Would you prefer to have more samples?

NOTE about approach: For the following rejection sampling, my approach was: my proposal distribution was just essentially a rectangle composed of two uniform distributions: one which samples the cheating rate from 0 to 1, and the other which samples the probability density from 0 to 0.05 (which is higher than the highest point on my posterior distribution). My target distribution is just my posterior, which - for each theta chosen from the first uniform distribution, I can then calculate the associated density by 1) finding the likelihood associated with 40/100 for a binomial with the correct coresponding probability (0.25 + 0.5 * theta) and multiplying that by the prior for that theta (to get the posterior associated with that theta), and then 2) seeing if the probability density sampled from my second distribution is above or below that calculated likelihood; if it's below, then the sample is kept, if it's above, then the sample is rejected.

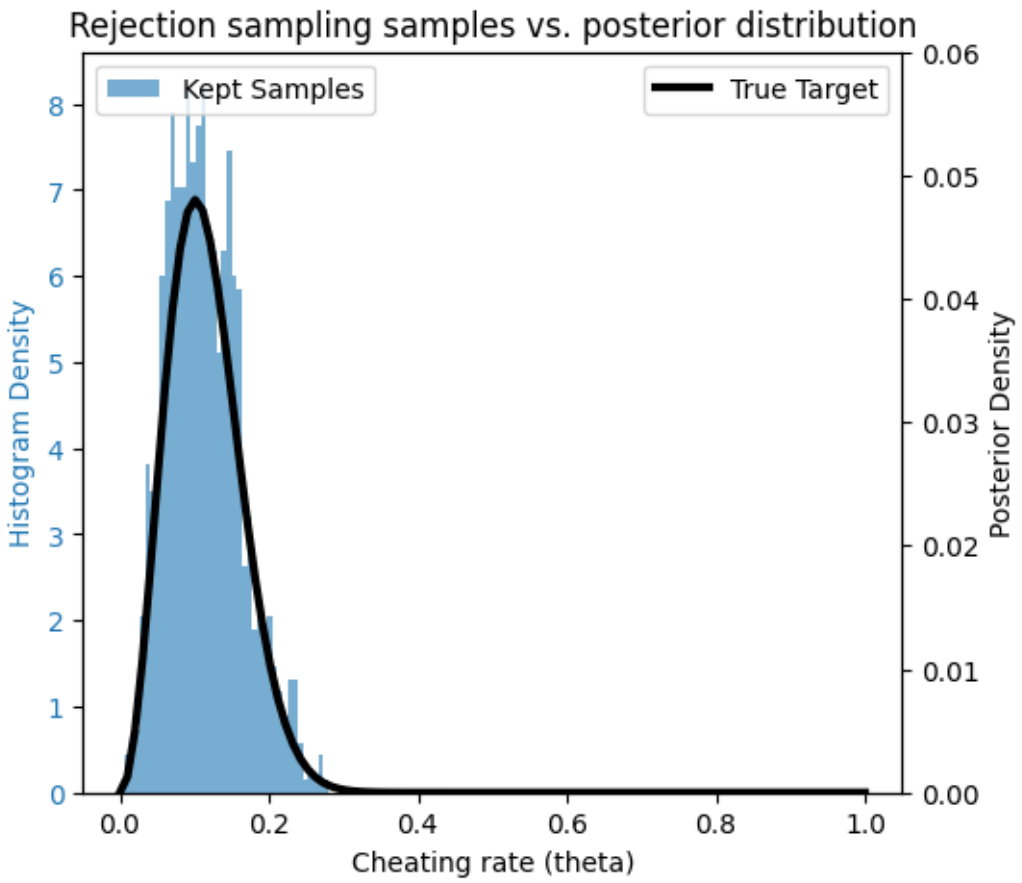
[10]:
```
# # Rejection sampling
# nsamp = 1_000
# unif_theta = np.random.uniform(0, 1, nsamp)
```

```
# unif_density = np.random.uniform(0, 0.07, nsamp)

# keep_samp = []
# for i in np.arange(0, len(unif_theta)):
#     likelihood_ = find_likelihood(unif_theta[i])
#     #print('likelihood_', likelihood_)
#     prior_ = find_prior(unif_theta[i])
#     #print('prior_', prior_)
#     posterior_ = np.exp(np.log(likelihood_)+np.log(prior_))
#     #print('posterior_',posterior_)
#     # check if posterior is above or below the density sampled from the␣
 ↪second unif dist; if below - keep:
#     #print(unif_density[i])
#     if unif_density[i]< posterior_:
#         #print('keep sample')
#         keep_samp.append(unif_theta[i])
# print(f'The number of samples kept out of {nsamp}, is {len(keep_samp)}')
```

[11]:
```
# Rejection sampling
random.seed(44)
nsamp = 1_000

keep_samp = []
num_samp_kep = 0

# run loop until 1,000 samples are kept via rejection sampling
while num_samp_kep < nsamp:
    unif_theta = np.random.uniform(0, 1, 1)
    unif_density = np.random.uniform(0, 0.07, 1)
    likelihood_ = find_likelihood(unif_theta)
    #print('likelihood_', likelihood_)
    prior_ = find_prior(unif_theta)
    #print('prior_', prior_)
    posterior_ = np.exp(np.log(likelihood_)+np.log(prior_))
    #print('posterior_',posterior_)
    # check if posterior is above or below the density sampled from the second␣
 ↪unif dist; if below - keep:
    #print(unif_density[i])
    if unif_density< posterior_:
        #print('keep sample')
        keep_samp.append(unif_theta[0])
        num_samp_kep = len(keep_samp)
print(f'The number of samples kept out of {nsamp}, is {len(keep_samp)}')
```

```
The number of samples kept out of 1000, is 1000
```

[12]:
```
fig, ax1 = plt.subplots(figsize=(5.5, 5))

# Plot histogram on primary y-axis
ax1.hist(keep_samp, bins=40, density=True, alpha=0.6, label='Kept Samples')#,␣
 ↪color='C0')
ax1.set_xlabel("Cheating rate (theta)")
ax1.set_ylabel("Histogram Density", color='C0')
ax1.tick_params(axis='y', labelcolor='C0')

# Create a second y-axis
ax2 = ax1.twinx()
ax2.plot(df['theta'], df['posterior'], linewidth=3, color='black', label='True␣
 ↪Target')
ax2.set_ylabel("Posterior Density", color='black')
ax2.tick_params(axis='y', labelcolor='black')
ax2.set_ylim(0, 0.06)

# Add legends
ax1.legend(loc='upper left')
ax2.legend(loc='upper right')

plt.title('Rejection sampling samples vs. posterior distribution')

plt.show()
```

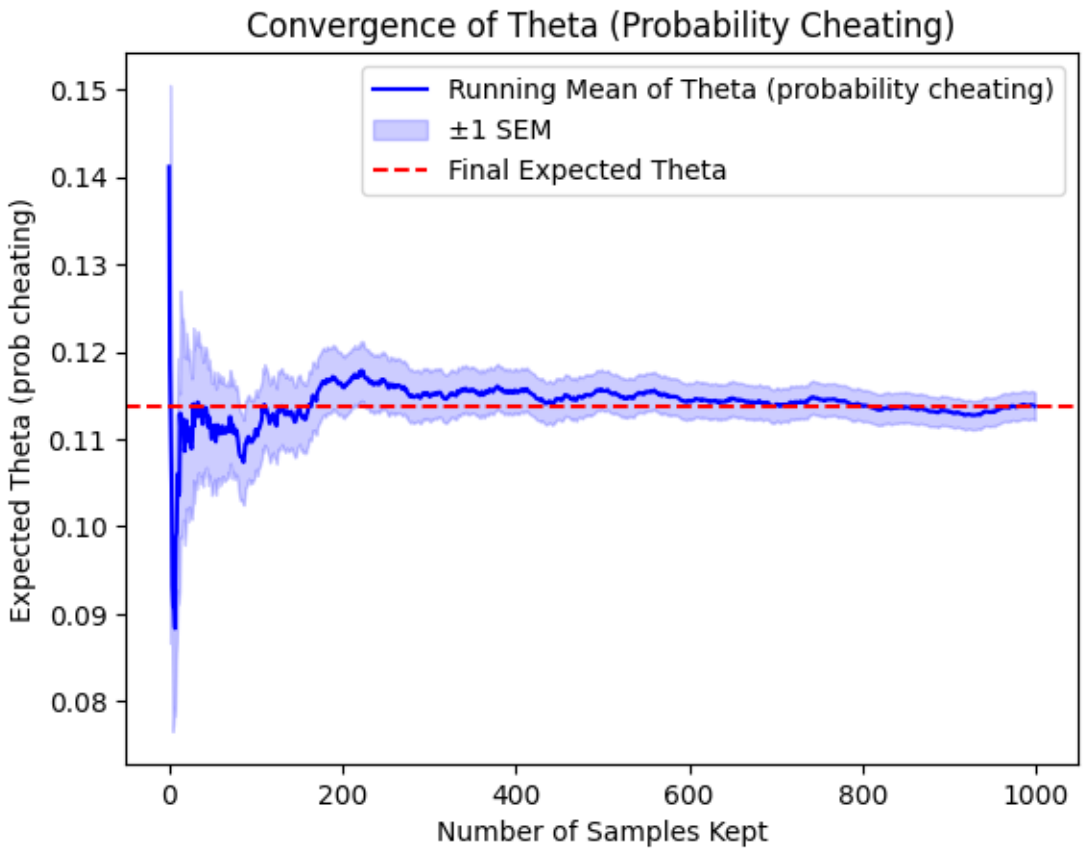


^Note that because I'm using dual y-axes the comparison looks a little funky, but overall reflects that samples align with posterior distribution.

[13]:
```
theta_running = pd.DataFrame({'theta_sampled': keep_samp})
theta_running['sim'] = np.arange(len(theta_running))

# Compute running mean and standard deviation of NPV over simulations
theta_running['theta_mean'] = theta_running['theta_sampled'].expanding().mean()
theta_running['theta_mean_std'] = theta_running['theta_sampled'].expanding().
 ↪std()

# Compute running standard error (SEM)
theta_running['theta_sem'] = theta_running['theta_mean_std'] / np.
 ↪sqrt(theta_running['sim'])  # sqrt of sample size

# theta_running
```

[14]:
```
# Plot running mean
```

```
plt.plot(theta_running['sim'], theta_running['theta_mean'], label='Running Mean␣
 ↪of Theta (probability cheating)', color='blue')

# confidnence int - plus minus 1 standard error in mean
plt.fill_between(theta_running['sim'],
                 theta_running['theta_mean'] - theta_running['theta_sem'],
                 theta_running['theta_mean'] + theta_running['theta_sem'],
                 color='blue', alpha=0.2, label='±1 SEM')

# Horizontal line for final expected NPV
plt.axhline(y=theta_running['theta_mean'].iloc[-1], color='red',␣
 ↪linestyle='--', label='Final Expected Theta')

# Labels and title
plt.xlabel("Number of Samples Kept")
plt.ylabel("Expected Theta (prob cheating)")
plt.legend()
plt.title(f"Convergence of Theta (Probability Cheating)")
plt.show()
```



Commentary RE: How well are you capturing the posterior you obtained from the grid approximation? Would you prefer to have more samples?

Quick note - initially I read this problem as do rejection sampling 1,000 times (not do rejection sampling until you ahve 1,000 samples from the posterior) - and was actually surprised at how quickly I thought the value converged (after ~100 samples) - now that I've redone this to have 1000 samples kept, I'm realizing it definitely hadn't converged after 100 samples (unsurprisingly) and it's made me more wary of assessing "convergence" in these plots. But - answering the main question - I think thes 1000 samples from my posterior obtained via rejection sampling are capturing my posterior reasonably well. It appears to me as though the expected value of theta has somewhat converged (based on the graph above) - but like I was saying originally, I'm hesitant to assume this has truly "converged" and would probably prefer to run it for ~10,000 samples (especially since it's not computationally expensive). That being said, the 1,000 samples are capturing the posterior relatively well (based on the comarison of the histogram and the pdf); and these samples are likely sufficient to get a decent estimate of the expected theta, and can be examined in combination with the calculated standard error in the estimate (shown via shading above). The 90% credible interval is discussed below.

D) Report the expected value of θ and its 90% credible interval. What conclusions can you draw about cheating probability and whether your belief in your class was warranted? How much uncertainty exists as a result of this privacy-protecting procedure?

[15]:
```
print(theta_running.tail(1))
```

```
     theta_sampled  sim  theta_mean  theta_mean_std  theta_sem
999       0.040653  999    0.113774        0.050131   0.001586
```

[16]:
```
theta_expected = np.mean(keep_samp)
theta_std = np.std(keep_samp)
theta_05, theta_95 = np.quantile(keep_samp, [0.05, 0.95])
theta_SEM = theta_std / np.sqrt(len(theta_running))
print(f'The expected theta is {round(theta_expected,2)}, with a standard error␣
 ↪in the estimate of the mean of {round(theta_SEM,3)}, and the 90% credible␣
 ↪interval for theta is {round(theta_05, 2)} to {round(theta_95,2)}.')
```

```
The expected theta is 0.11, with a standard error in the estimate of the mean of
0.002, and the 90% credible interval for theta is 0.04 to 0.21.
```

COMMENTARY - What conclusions can you draw about cheating probability and whether your belief in your class was warranted? How much uncertainty exists as a result of this privacy-protecting procedure?: Based on the analyis above, the expected probability of cheating is approximately 11% and the 90% credible interval is approximately 4 to 21%. As stated in the question framing, our prior belief is that the class is not broadly engaged in cheating, and as such we used a relatively tightly constrained prior. I think that to some extent, this tightly constraining prior heavily dictated the shape of the posterior (as can be seen by the way the prior "pulls" the posterior to the left, compared to the likelihood; explored more in the following quesiton). I think a 90% credible interval of 4 to 21% is too large to confidently conclude that cheating is not prevalent (i.e., 21% is pretty high in my opinion) - though I think this is largely dependent on the definition used for "prevalent cheating". I would say quite a bit of uncertainty still exists as a result of this privacy-protecting procedure, and it's difficult to draw meaningful

conclusions on the data we have here. I also think it's important to note that the tight prior is likely making this estimate appear less uncertain than it actually is (unless we had really good reason to believe that our prior belief is valid and should influence the results this heavily).

E) Repeat the analysis with a less constrained prior of your choosing. How does that change your inferences and conclusions? What does this tell you about the effectiveness of the procedure?

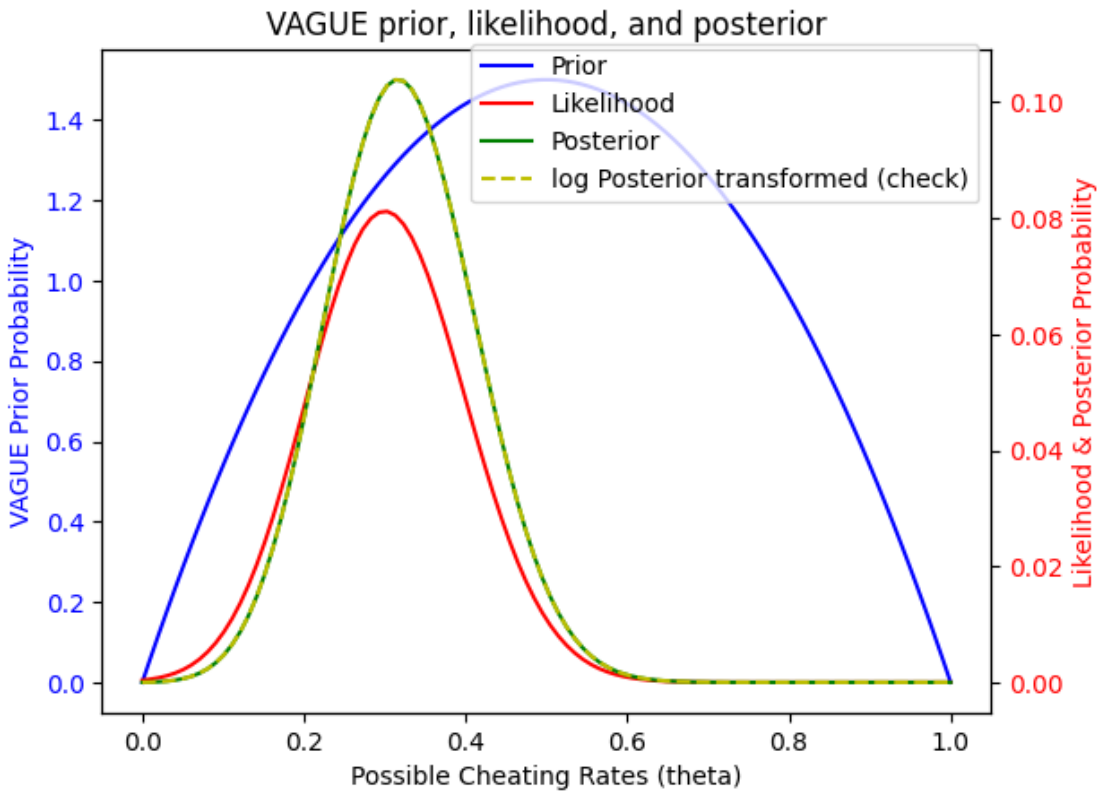
[17]:
```
def find_prior_vague(theta_):
    # x, a, b, loc=0, scale=1
    prior_ = stats.beta.pdf(theta_, a = 2, b = 2, loc=0, scale=1)
    return prior_

# num_yes_cheat = np.arange(1, 101)
beta_prior_vague = []
for i in np.arange(0, len(theta)):
    prior_ = find_prior_vague(theta[i])
    beta_prior_vague.append(prior_)
```

[18]:
```
df_vague_prior = pd.DataFrame({
    'theta': theta,
    'likelihood': likelihood,
    'beta_prior': beta_prior_vague
})

df_vague_prior['posterior'] =␣
 ↪df_vague_prior['likelihood']*df_vague_prior['beta_prior']

df_vague_prior['log_likelihood'] = np.log(df_vague_prior['likelihood'])
df_vague_prior['log_beta_prior'] = np.log(df_vague_prior['beta_prior'])

df_vague_prior['log_posterior'] =␣
 ↪df_vague_prior['log_likelihood']+df_vague_prior['log_beta_prior']

df_vague_prior['log_posterior_transform'] = np.
 ↪exp(df_vague_prior['log_posterior'])
```

```
/Users/mca76/Documents/Courses/EnvDataAnalysis/HW/bee5850_python_env/lib/python3
.11/site-packages/pandas/core/arraylike.py:399: RuntimeWarning: divide by zero
encountered in log
  result = getattr(ufunc, method)(*inputs, **kwargs)
```

[19]:
```
fig, ax1 = plt.subplots()

# Primary y-axis (Prior)
ax1.plot(df_vague_prior['theta'], df_vague_prior['beta_prior'], 'b-',␣
 ↪label='Prior')
ax1.set_xlabel('Possible Cheating Rates (theta)')
ax1.set_ylabel('VAGUE Prior Probability', color='b')
```

```
ax1.tick_params(axis='y', labelcolor='b')

# Secondary y-axis (Likelihood & Posterior)
ax2 = ax1.twinx()
ax2.plot(df_vague_prior['theta'], df_vague_prior['likelihood'], 'r-',␣
 ↪label='Likelihood')
ax2.plot(df_vague_prior['theta'], df_vague_prior['posterior'], 'g-',␣
 ↪label='Posterior')
ax2.plot(df_vague_prior['theta'], np.exp(df_vague_prior['log_posterior']),␣
 ↪'y--', label='log Posterior transformed (check)')
ax2.set_ylabel('Likelihood & Posterior Probability', color='r')
ax2.tick_params(axis='y', labelcolor='r')

# Title and Legend
plt.title('VAGUE prior, likelihood, and posterior')
fig.legend(loc='upper right', bbox_to_anchor=(.9,.9))

plt.show()
```



^ The posterior probability is plotted in green (obtained via multiplying liklihood and prior) and yellow (obtained via adding log likelihood and log prior, then exponentiating) above. Note that

the scales of the prior, likelihood, and posterior are a bit wonky (and not really meaningful/do not impact the analysis) - to make this visually a bit nicer I could have normalized each to their respective max densities, but I didn't bother (based on conversation with Vivek during office hours).

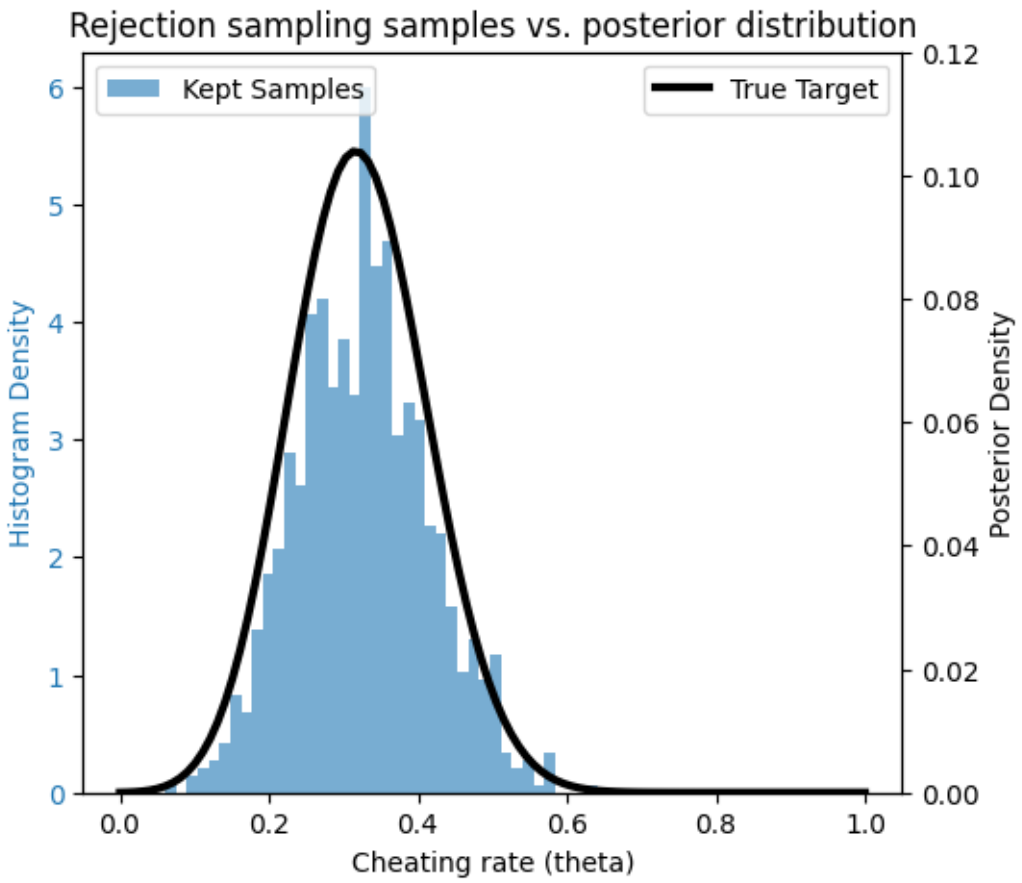
[20]:
```
# # Rejection sampling
# nsamp = 1_000
# unif_theta = np.random.uniform(0, 1, nsamp)
# unif_density = np.random.uniform(0, 0.11, nsamp)

# keep_samp = []
# for i in np.arange(0, len(unif_theta)):
#     likelihood_ = find_likelihood(unif_theta[i])
#     #print('likelihood_', likelihood_)
#     prior_ = find_prior_vague(unif_theta[i])
#     #print('prior_', prior_)
#     posterior_ = np.exp(np.log(likelihood_)+np.log(prior_))
#     #print('posterior_',posterior_)
#     # check if posterior is above or below the density sampled from the␣
 ↪second unif dist; if below - keep:
#     #print(unif_density[i])
#     if unif_density[i]< posterior_:
#         #print('keep sample')
#         keep_samp.append(unif_theta[i])
# print(f'The number of samples kept out of {nsamp}, is {len(keep_samp)}')
```

[21]:
```
# Rejection sampling
random.seed(44)
nsamp = 1_000

keep_samp_vague = []
num_samp_kep = 0

# run loop until 1,000 samples are kept via rejection sampling
while num_samp_kep < nsamp:
    unif_theta = np.random.uniform(0, 1, 1)
    unif_density = np.random.uniform(0, 0.11, 1)
    likelihood_ = find_likelihood(unif_theta)
    #print('likelihood_', likelihood_)
    prior_ = find_prior_vague(unif_theta)
    #print('prior_', prior_)
    posterior_ = np.exp(np.log(likelihood_)+np.log(prior_))
    #print('posterior_',posterior_)
    # check if posterior is above or below the density sampled from the second␣
 ↪unif dist; if below - keep:
    #print(unif_density[i])
    if unif_density< posterior_:
        #print('keep sample')
```

```
        keep_samp_vague.append(unif_theta[0])
        num_samp_kep = len(keep_samp_vague)
print(f'The number of samples kept out of {nsamp}, is {len(keep_samp_vague)}')
```

```
The number of samples kept out of 1000, is 1000
```

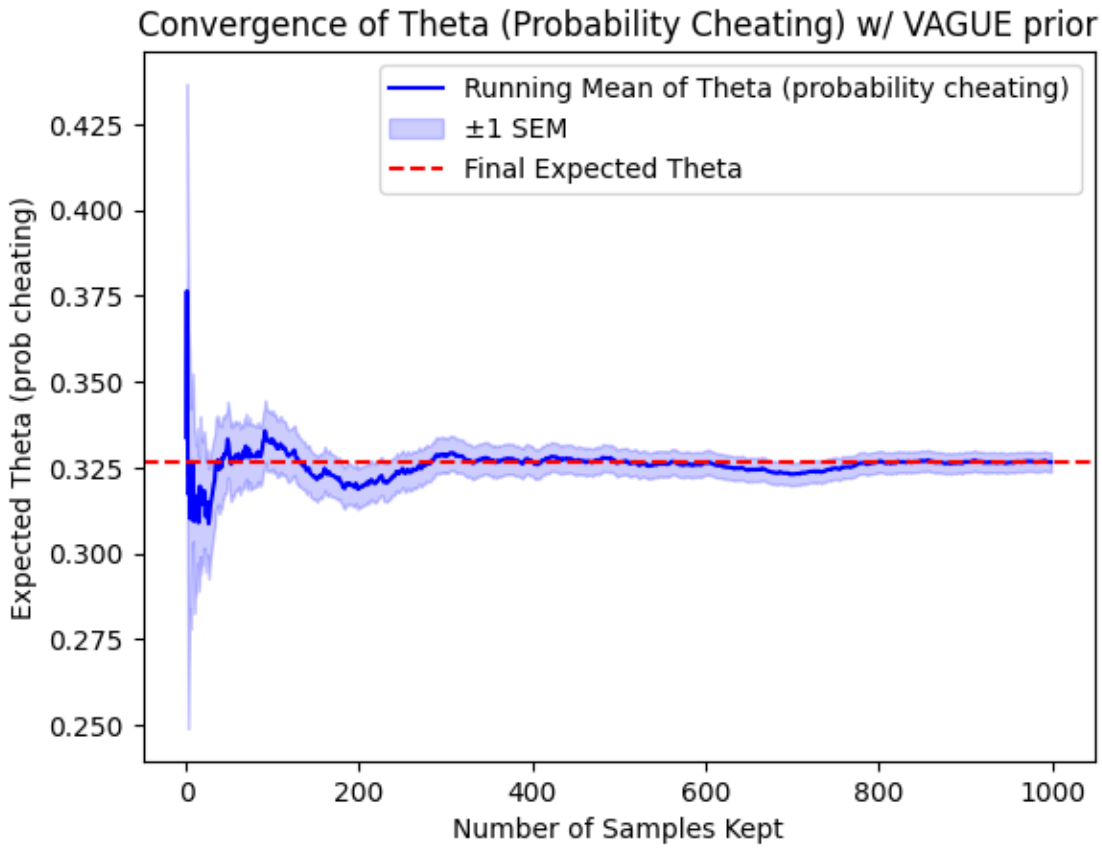
[22]:
```
fig, ax1 = plt.subplots(figsize=(5.5, 5))

# Plot histogram on primary y-axis
ax1.hist(keep_samp_vague, bins=40, density=True, alpha=0.6, label='Kept␣
 ↪Samples', color='C0')
ax1.set_xlabel("Cheating rate (theta)")
ax1.set_ylabel("Histogram Density", color='C0')
ax1.tick_params(axis='y', labelcolor='C0')

# Create a second y-axis
ax2 = ax1.twinx()
ax2.plot(df_vague_prior['theta'], df_vague_prior['posterior'], linewidth=3,␣
 ↪color='black', label='True Target')
ax2.set_ylabel("Posterior Density", color='black')
ax2.tick_params(axis='y', labelcolor='black')
ax2.set_ylim(0, 0.12)

# Add legends
ax1.legend(loc='upper left')
ax2.legend(loc='upper right')

plt.title('Rejection sampling samples vs. posterior distribution')

plt.show()
```



[23]:
```
theta_running_vague = pd.DataFrame({'theta_sampled': keep_samp_vague})
theta_running_vague['sim'] = np.arange(len(theta_running_vague))

# Compute running mean and standard deviation of NPV over simulations
theta_running_vague['theta_mean'] = theta_running_vague['theta_sampled'].
 ↪expanding().mean()
theta_running_vague['theta_mean_std'] = theta_running_vague['theta_sampled'].
 ↪expanding().std()

# Compute running standard error (SEM)
theta_running_vague['theta_sem'] = theta_running_vague['theta_mean_std'] / np.
 ↪sqrt(theta_running_vague['sim'])  # sqrt of sample size

# theta_running_vague
```

[24]:
```
# Plot running mean
plt.plot(theta_running_vague['sim'], theta_running_vague['theta_mean'],␣
 ↪label='Running Mean of Theta (probability cheating)', color='blue')
```

```
# confidnence int - plus minus 1 standard error in mean
plt.fill_between(theta_running_vague['sim'],
                 theta_running_vague['theta_mean'] -␣
 ↪theta_running_vague['theta_sem'],
                 theta_running_vague['theta_mean'] +␣
 ↪theta_running_vague['theta_sem'],
                 color='blue', alpha=0.2, label='±1 SEM')

# Horizontal line for final expected NPV
plt.axhline(y=theta_running_vague['theta_mean'].iloc[-1], color='red',␣
 ↪linestyle='--', label='Final Expected Theta')

# Labels and title
plt.xlabel("Number of Samples Kept")
plt.ylabel("Expected Theta (prob cheating)")
plt.legend()
plt.title(f"Convergence of Theta (Probability Cheating) w/ VAGUE prior")
plt.show()
```



[25]:
```
theta_expected = np.mean(keep_samp_vague)
theta_std = np.std(keep_samp_vague)
theta_05, theta_95 = np.quantile(keep_samp_vague, [0.05, 0.95])
theta_SEM = theta_std / np.sqrt(len(theta_running))
print(f'The expected theta using a VAGUE PRIOR is {round(theta_expected,2)},␣
 ↪with a standard error in the mean of {round(theta_SEM,3)}, and the 90%␣
 ↪credible interval for theta is {round(theta_05, 2)} to {round(theta_95,2)}.')
```

```
The expected theta using a VAGUE PRIOR is 0.33, with a standard error in the
mean of 0.003, and the 90% credible interval for theta is 0.19 to 0.48.
```

COMMENTARY RE: INFERENCES & CONCLUSIONS W VAGUE PRIOR: In the section above, I repeated the anlaysis with a less constrained prior, and got markedly different results. Unlike the constrained prior that heavily shifted our posterior away from the likelihood, this vague prior allows the posterior to have a shape that is a lot more similar to the likelihood (as shown above). The resulting estiamtes of theta are very different - the estimated theta is 0.33, with a 90% credible interval of 0.19 to 0.48, which almost entirely does not overlap my 90% credible interval from the original quesiton!! Based on these modified results (w/ vague prior), I would conclude that it is much more likely that cheating may be prevalent. E.g., if 15% cheating would be considered unacceptably "prevalent", then my results form this second analysis suggest that cheating is in fact prevalent, whereas my results from the original problem would suggest that it's likely not prevalent, but still could not be ruled out as being prevalent since the 90% credible interval includes 0.15. **Clearly, the choice of prior can dramatically impact inferences and conclusions**, and is a critical step in this modeling process (and a tightly constraining prior should only be used if you have very solid evidence for why this prior belief should limit results in your analysis). **Either way - there is a lot of uncertainty in our estimate of theta with this coin flipping approach** (showcased by the large 90% credible interval in both questions) - **suggesting the effectiveness of the coin flipping procedure is relatively low.**

1.3.3 Problem 2

You have been asked by a client to assess the risks of flooding to their home (which is valued at $400,000 and only floods when the stream water levels exceed 1.5m). They would like to know the probability distribution of annual flood damages to their home if they do not take any preventative floodproofing measures. After some hydrologic analysis, you have developed a 40-year record of annual maximum water levels (in m) at the nearby stream.

You also have a depth-damage function for the fraction of the home's value which is damaged at varying flood depths:

$$d(h) = \mathbb{I}[h > 0]\frac{1}{1 + \exp(-k(x - x_0))},$$

where $k = 1.25$ and $x_0 = 2$[1]. The graph of this function is given in the figure below.

[1] As a reminder, the indicator function $\mathbb{I}[h > 0]$ is 0 when the condition is not satisfied (in this case, when $h = 0$) and 1 when it is satisfied.

In this problem:

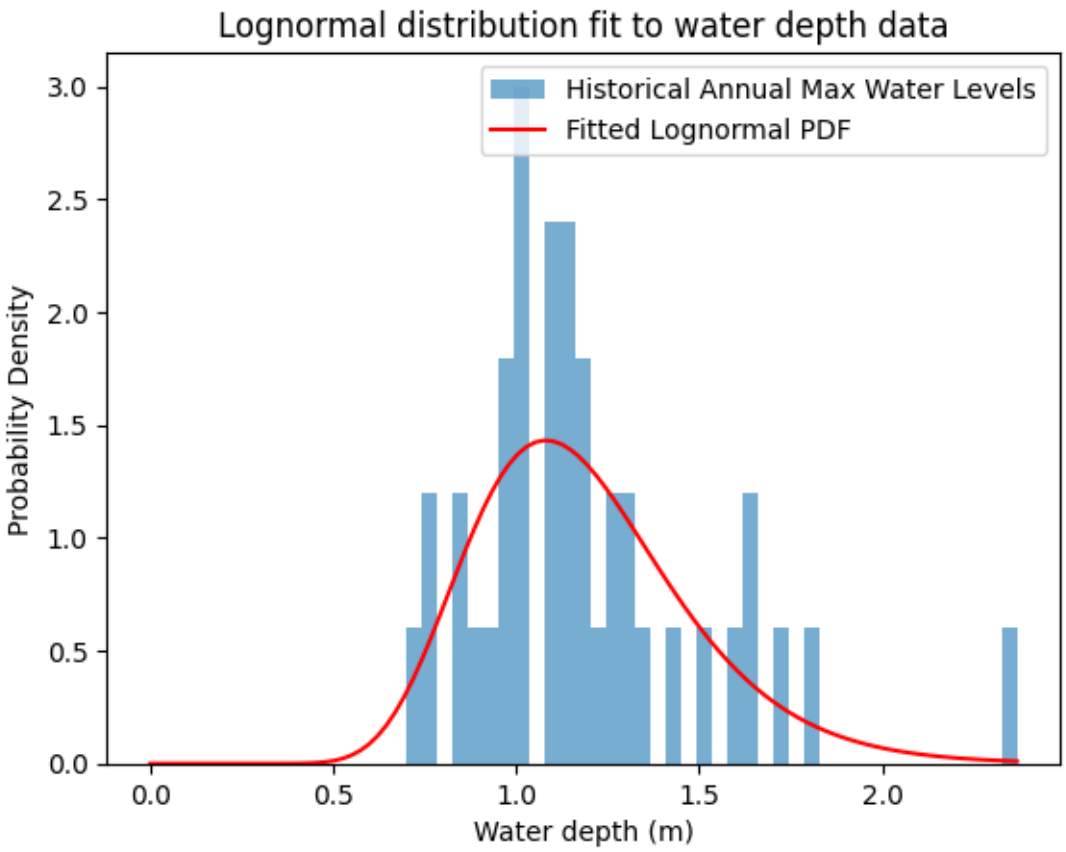
- Fit a log-normal distribution to the water depth data by maximizing likelihood.

- Use 10,000 samples from this distribution to estimate the expected damages from an annual maximum flood event using Monte Carlo simulation. Report your estimate as well as its standard error. Would you want to use more samples? Why or why not?

USED CHATGPT (OpenAI, 2025): converted Julia to Python to convert course notes code on rejection sampling to python; any additional changes reflect my own work. Also used chatgpt to troubleshoot an error I was getting when I tried to multiply damage fractions by 400,000, turns out you can't multiply a list of values by a number haha so chatgpt told me to convert to numpy array first which fixed the problem i was having.

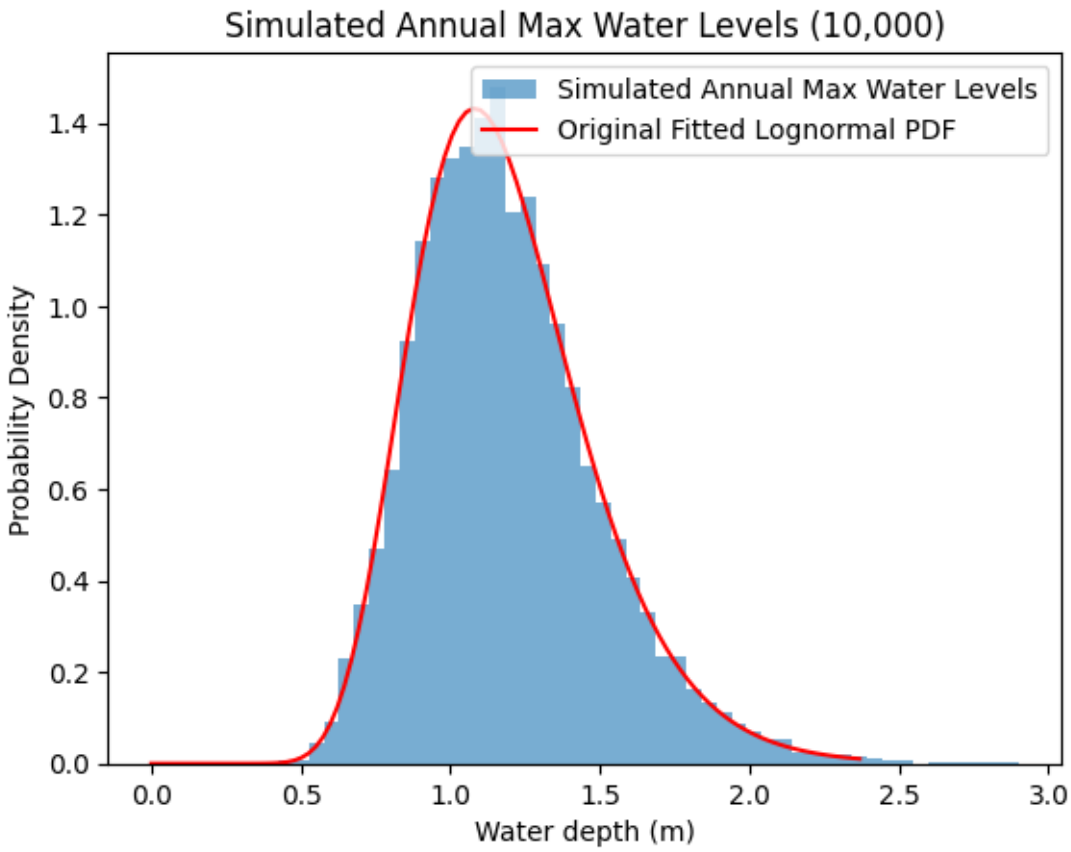
[26]:
```
# read in data:
water = pd.read_csv("data/water_depths.csv")
# print(water.head())

# log-normal distribution to the water depth data by maximizing likelihood:
shape, loc, scale = stats.lognorm.fit(water['depths'], floc= 0)
print(f'MLE outputs of lognormal dist: shape {shape}, loc {loc}, scale {scale}')



# get pdf and plot
x = np.linspace(0, max(water['depths']), 100) #min(water['depths'])
# pdf(x, s, loc=0, scale=1)
pdf = stats.lognorm.pdf(x, shape, loc, scale)
plt.hist(water, bins=40, density=True, alpha=0.6, label='Historical Annual Max␣
 ↪Water Levels')
plt.plot(x, pdf, 'r-', label='Fitted Lognormal PDF')
plt.xlabel('Water depth (m)')
plt.ylabel('Probability Density')
plt.title('Lognormal distribution fit to water depth data')
plt.legend()
plt.show()
```

```
MLE outputs of lognormal dist: shape 0.24963728245631378, loc 0, scale
1.1509354338959927
```



Note that the algorithm used to fit the lognormal distribution above is the version that will ultimately lead to all negative net benefit-cost calculations (as discussed in ed discussion post).

[27]:
```
random.seed(44)
# 10,000 samples from distribution
# take 10,000 random samples from this lognormal pdf,
samples = stats.lognorm.rvs(s = shape, loc=loc, scale=scale, size=10000,␣
 ↪random_state=None)

# plot samples for reference:
plt.hist(samples, bins=50, density=True, alpha=0.6, label='Simulated Annual Max␣
 ↪Water Levels')
plt.plot(x, pdf, 'r-', label='Original Fitted Lognormal PDF')
plt.xlabel('Water depth (m)')
plt.ylabel('Probability Density')
plt.title('Simulated Annual Max Water Levels (10,000)')
plt.legend()
plt.show()
```



Now, using the depth-damage function for the fraction of the home's value which is damaged at varying flood depths:

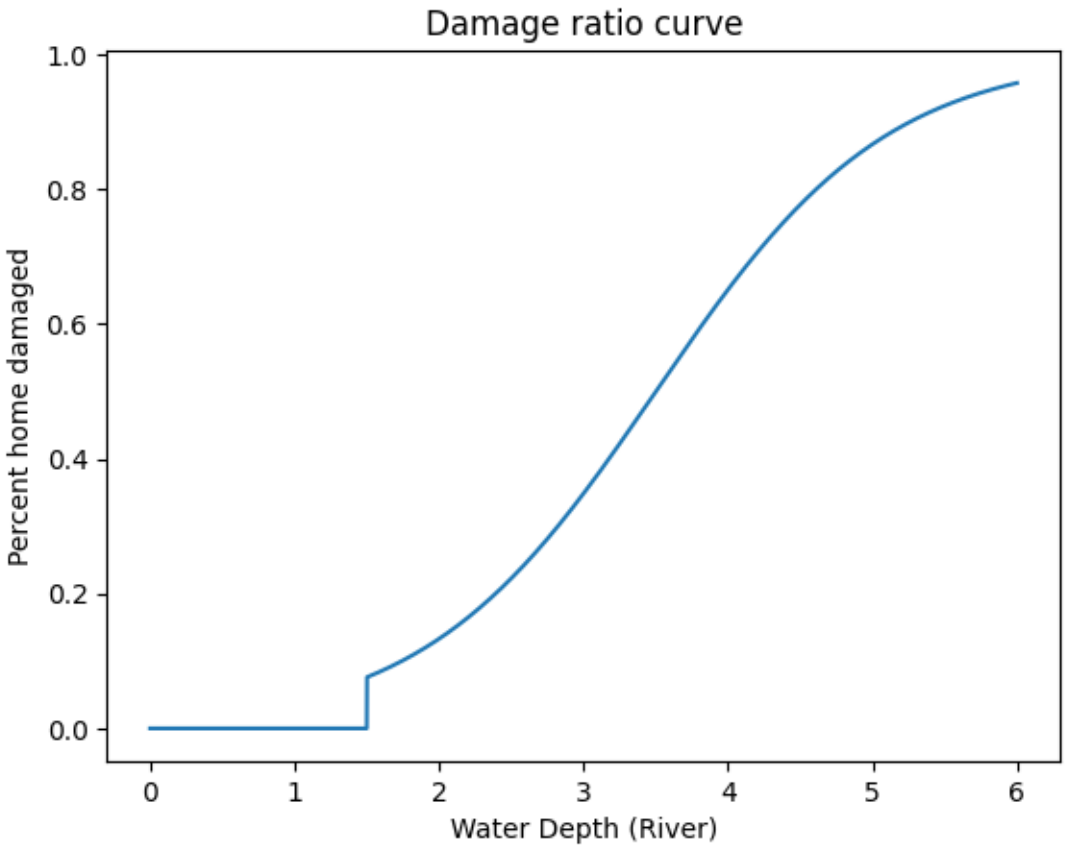
$$d(h) = \mathbb{I}[h > 0]\frac{1}{1 + \exp(-k(x - x_0))},$$

where $k = 1.25$ and $x_0 = 2$[1].

NOTE ABOUT TACTIC: In the problem framing, the house only floods when the height of the water is 1.5m or higher. As such, I first calculated h from the water depth above (i.e., water depth from simulation - 1.5), and then proceeded to use the equation where damage occurs only if h>0, i.e., if water depth is > 1.5. In this framing, the x in in the equation is also h (i.e., depth of water in home). Additionally, we know the original value of the house is $400,000, which we can use to translate fraction of value damaged into an actual dollar amount.

[28]:
```
# checkign that i have correct curve:
k = 1.25
x_0 = 2
grid_depths = np.linspace(0,6,10000)
damage_ratio = []
for i in range(0,len(grid_depths)):
```

```
    h = grid_depths-1.5
    damage = 0
    if h[i] > 0:
        damage = (1/(1+np.exp(-k*(h[i]-x_0))))
    damage_ratio.append(damage)

plt.plot(grid_depths, damage_ratio)
plt.title('Damage ratio curve')
plt.xlabel('Water Depth (River)')
plt.ylabel('Percent home damaged')
plt.show()
```



[29]:
```
# estimate the expected damages from an annual maximum flood event using Monte␣
 ↪Carlo simulation
# i.e., plug in water level sample estimate to the equation above
h = samples-1.5
num_exceed_thresh = (h>0).sum()

k = 1.25
```

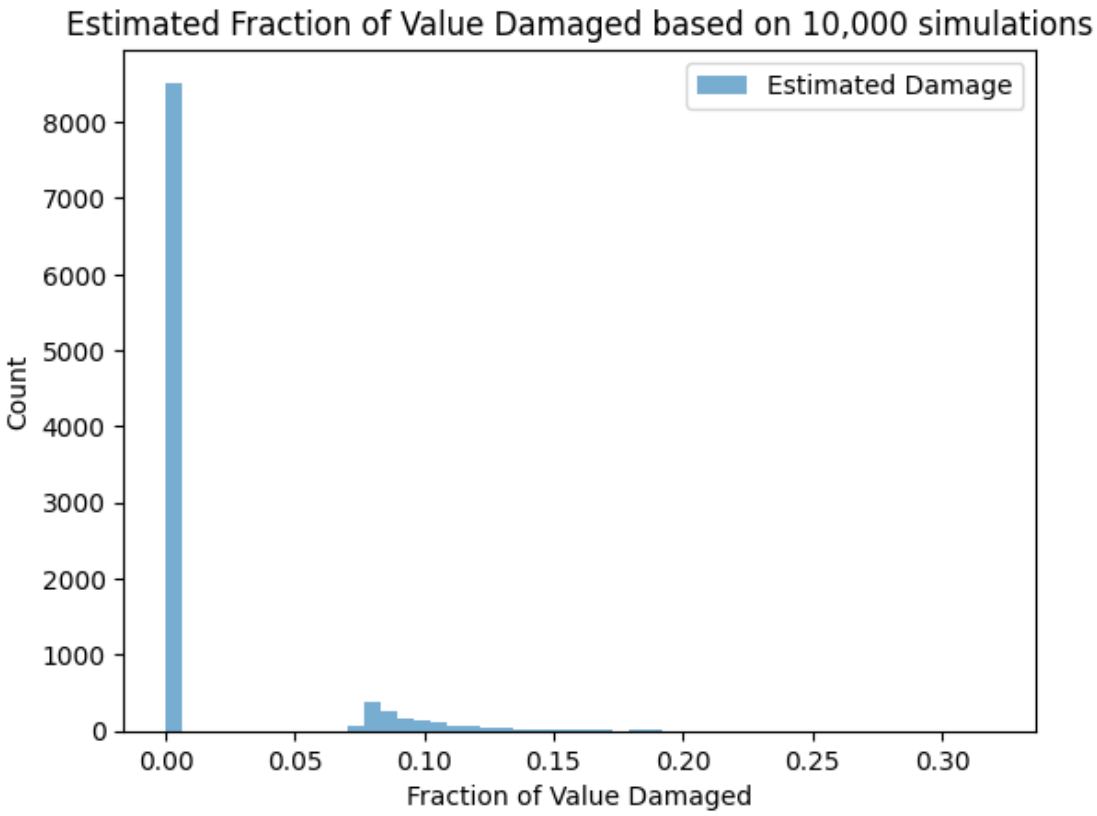
```
x_0 = 2
damages = []
for i in range(0,len(h)):
    damage = 0
    if h[i] > 0:
        damage = (1/(1+np.exp(-k*(h[i]-x_0))))
    damages.append(damage)
damages = np.array(damages)
damages_dollars = (damages*400000)
```

[30]:
```
print(f'The number of simulated events exceeding the 1.5 threshold is:␣
↪{num_exceed_thresh}, and confirming that damages were calculated for:␣
↪{len(damages)}.')
```

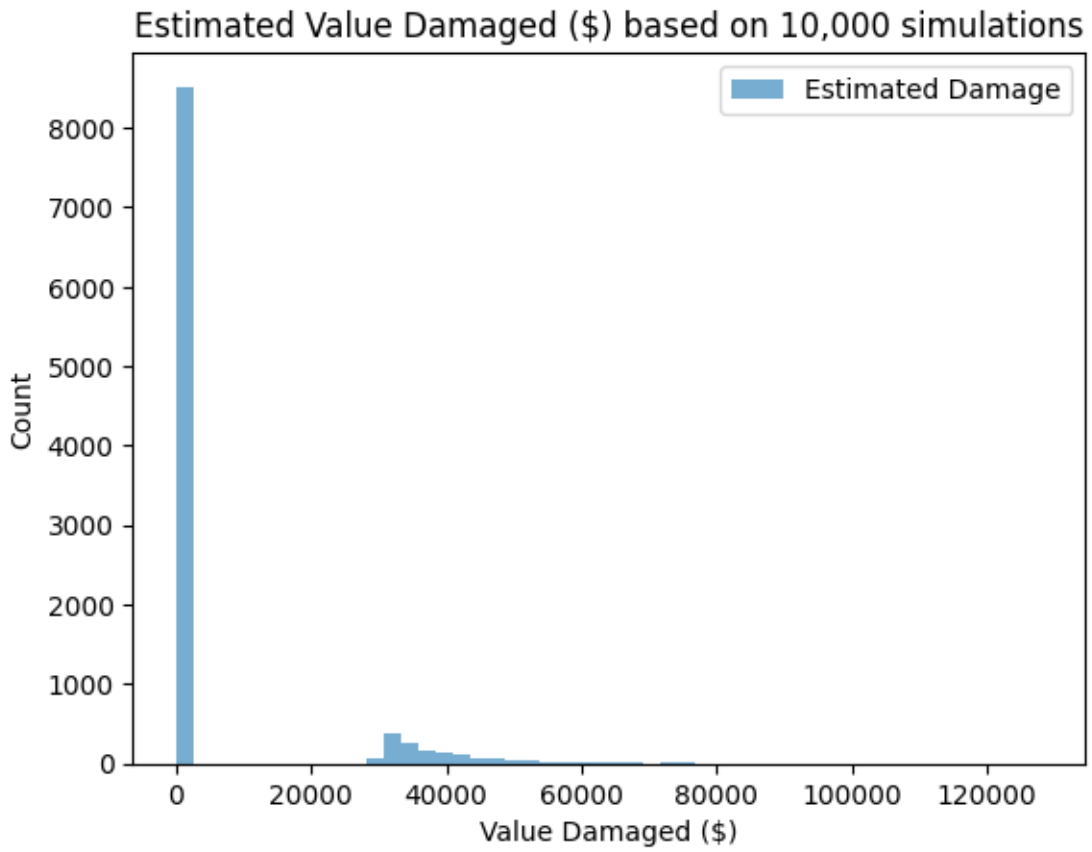
```
The number of simulated events exceeding the 1.5 threshold is: 1481, and
confirming that damages were calculated for: 10000.
```

NOW: Use 10,000 samples from this distribution to estimate the expected damages from an annual maximum flood event using Monte Carlo simulation.

[31]:
```
plt.hist(damages, bins=50, density=False, alpha=0.6, label='Estimated Damage')
plt.xlabel('Fraction of Value Damaged')
plt.ylabel('Count')
plt.title('Estimated Fraction of Value Damaged based on 10,000 simulations')
plt.legend()
plt.show()
```



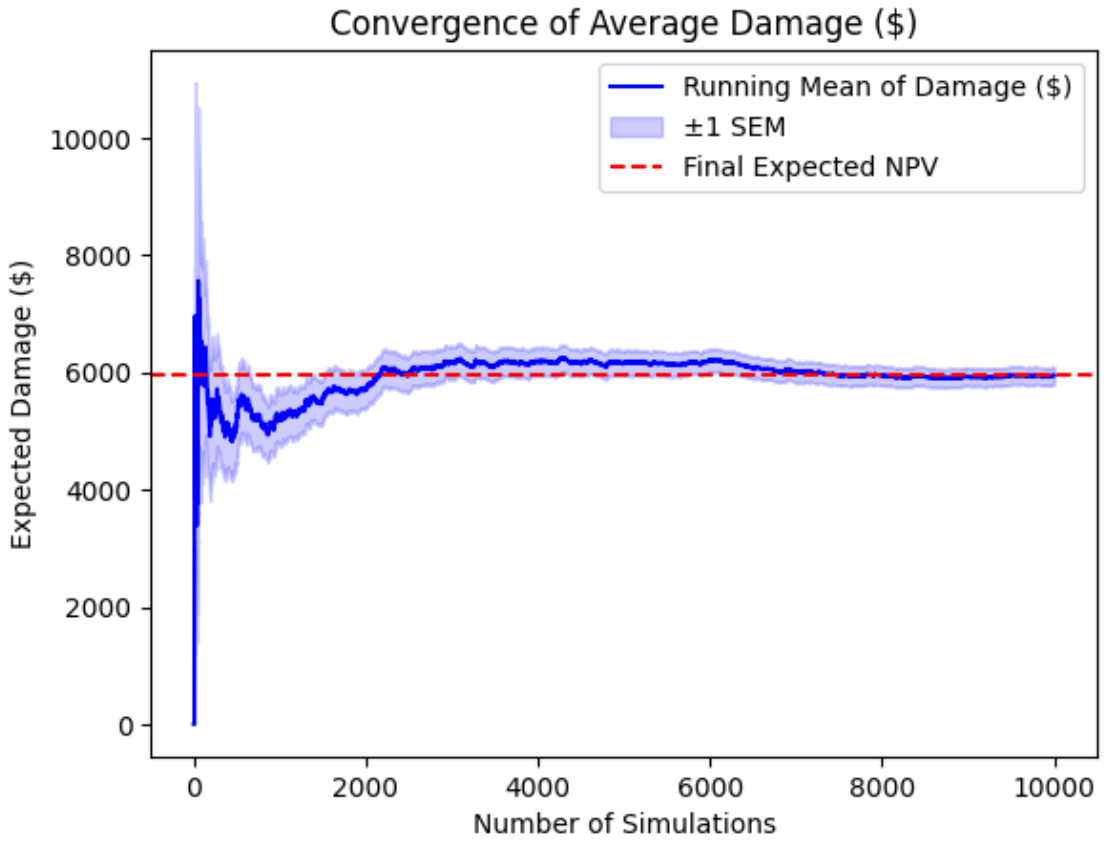
[33]:
```
plt.hist(damages_dollars, bins=50, density=False, alpha=0.6, label='Estimated␣
↪Damage')
plt.xlabel('Value Damaged ($)')
plt.ylabel('Count')
plt.title('Estimated Value Damaged ($) based on 10,000 simulations')
plt.legend()
plt.show()
```



[34]:
```
df_running = pd.DataFrame({'damages_dollars': damages_dollars})
df_running['sim'] = np.arange(len(df_running))

# Compute running mean and standard deviation of NPV over simulations
df_running['damages_dollars_mean'] = df_running['damages_dollars'].expanding().
 ↪mean()
df_running['damages_dollars_std'] = df_running['damages_dollars'].expanding().
 ↪std()

# Compute running standard error (SEM)
df_running['damages_dollars_sem'] = df_running['damages_dollars_std'] / np.
 ↪sqrt(df_running.index + 1)  # sqrt of sample size
```

[35]:
```
# Plot running mean
plt.plot(df_running['sim'], df_running['damages_dollars_mean'], label='Running␣
 ↪Mean of Damage ($)', color='blue')

# confidnence int - plus minus 1 standard error in mean
plt.fill_between(df_running['sim'],
```

```
                df_running['damages_dollars_mean'] -␣
 ↪df_running['damages_dollars_sem'],
                df_running['damages_dollars_mean'] +␣
 ↪df_running['damages_dollars_sem'],
                color='blue', alpha=0.2, label='±1 SEM')

# Horizontal line for final expected NPV
plt.axhline(y=df_running['damages_dollars_mean'].iloc[-1], color='red',␣
 ↪linestyle='--', label='Final Expected NPV')

# Labels and title
plt.xlabel("Number of Simulations")
plt.ylabel("Expected Damage ($)")
plt.legend()
plt.title(f"Convergence of Average Damage ($)")
plt.show()
```



[36]:
```
# calculating the damage estiamte mean, standard deviation, and standard error␣
 ↪in mean estiamte.
```

```
damage_estimate = np.mean(damages_dollars)
damage_std = np.std(damages_dollars)
damage_estimate_SEM = damage_std/np.sqrt(len(damages_dollars))
print(f'Estimated value damaged ($): {round(damage_estimate,2)} and estiamted␣
 ↪standard error ($): {round(damage_estimate_SEM,2)}')
```

```
Estimated value damaged ($): 5940.66 and estiamted standard error ($): 148.83
```

COMMENTARY - RE: Report your estimate as well as its standard error. Would you want to use more samples? Why or why not?

Using Monte Carlo simulation, the estimated expected damages from an annual maximum flood event is approximately $5941, with a standard error in the estimate of this mean (expected value) of $149. I would not want to use more samples, as can be seen in the figure above - the convergence of the MC simulations to an expected value occurs relatively quickly, and the estimate varies very little after 4000 or so simulations. The standard error in the estimate of the mean of course will continue to decrease, but this happens on the order of 1/sqrt(n), i.e., it would take a lotttt more samples to get the standard error in the estiamte of the mean to by significantly better than the current standard error ($147).

1.3.4 Problem 3

GRADED FOR 5850 STUDENTS ONLY

Next, your client would like advice on whether it would be cost-effective to elevate their home to reduce flood risks. To address this question, you will need to calculate the **net present value (NPV)** of flooding damages, which converts damages over time to a present value using a discount rate[1] For example, if your discount rate is 4%, a dollar next year is worth the equivalent of about 96 cents today. More generally, with a discount rate of γ (as a decimal), a dollar of benefits in t years is worth $\$(1-\gamma)^t$ today.

The NPV of a sequence of money x_t with discount rate γ is the sum of all of the discounted values, that is,

$$NPV = \sum_{t=0}^{T} x_t (1-\gamma)^t,$$

where T is the time horizon. We'll use a discount rate of 4% in this problem, which is typical for this type of problem, and a design horizon of 30 years.

For this problem, we will assume that the cost of elevating the house Δh m is $C(\Delta h) = \mathbb{I}[h > 0](100,000 + 2,000\Delta h)$.

[1] Discount rates reduce future monetary values to reflect the time-value of money; that is, you would rather have a bit less money today than none today and more next year, as you could save or invest that money. The actual choice of discount rates is the subject of much economic theory and plays an important role in environmental decision-making, particularly for multi-generational investments such as climate mitigation. For example, with a relatively large discount rate (such as 7%), all costs and benefits after a decade are effectively zero, which means it never appears cost-effective to e.g. reduce fossil fuel emissions.

Tip

Be careful about the quantity you're estimating. Calculating the benefits of elevating by Δh m requires calculating how the elevation *changes* the expected flooding damages over the design horizon *relative to a no-heightening baseline*.

In this problem:

- Using the extreme water level distribution from Problem 2, use Monte Carlo simulation to estimate the NPV of the benefits of elevation levels between 0 and 5m (you don't have to consider increments finer than 0.5m). What elevation heights pass the cost-benefit test? You can assume that the costs of elevation are all up front (that is, in year 0).
- What height (including a possible elevation of 0m) maximizes the NPV of net benefits (benefits - costs)?[1]

[1] This is an example of **Monte Carlo optimization** as you are using Monte Carlo simulation to estimate the optimization objective function.

USED CHATGPT (OpenAI, 2025): I wrote out the following code (now commented out in the block below) and was having trouble thinking through the best way to store the data in a dataframe the way I wanted to (with each column representing a delta h between 0 and 5, and with each row representing on 30 year simulation resulting NPV). I asked chatgpt to suggest the best way to store the data, and it gave me a more streamlined version of my code - specifically, redoing my NPV calculation to use matrix multiplication so I don't have to loop through the years 1 through 30 the way I was originally doing; since the suggested modifications made the code much cleaner, I proceeded with that code (following block) - hope that's ok! The suggested code uses most of the same logic and syntax that I originally used, just made it way cleaner and reduced the number of loops I needed haha. I then added comments to it, added the portion where I calculate cost and add that to the df, and ultiamtely calculate cost-benefit tradeoff to the code (chatgpt didn't provide any of that)

[37]:
```
# # NPV = sum from time 0 to time T (30 years) of xt(1-gamma)^t where gamma is␣
 ↪0.04, t is the year, and xt is the damages that year??
# # cost of elevating house is (100,000 + 2,000 delta h), and all cost occurs␣
 ↪in time 0
# # delta h varies between 0 and 5 by 0.5 increments
# # compare

# # for each 30 years:
# T = 30
# gamma = 0.04
# k = 1.25
# x_0 = 2
# # delta h to examine:
# delta_h = np.linspace(0, 5, 11)
# #number of simulations
# for sim in np.arange(0, 100):
#     #for each 30 year period, get random sample of water heights:
#     samples_MC = stats.lognorm.rvs(s = shape, loc=loc, scale=scale, size=T,␣
 ↪random_state=None)
```

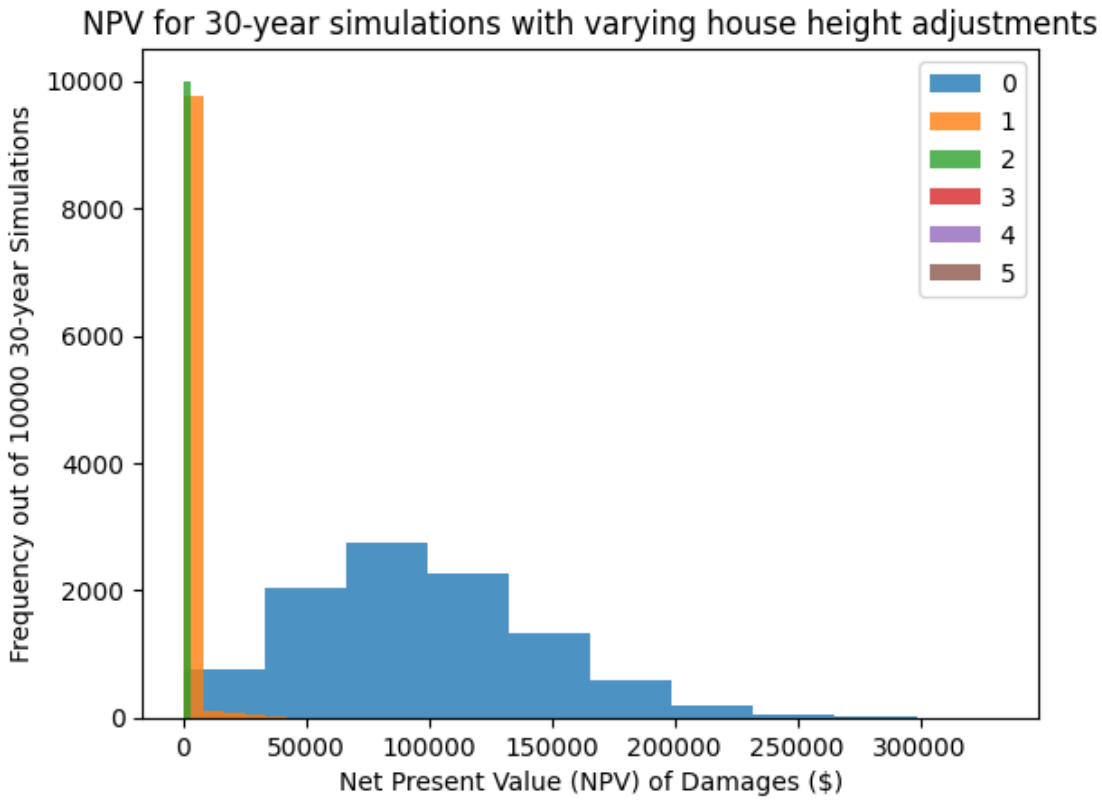
```
#     #Loop for different delta h values (different elevation amounts): 0 to 5m␣
 ↪by 0.5
#     for k in np.arange(0, len(delta_h)):
#         del_h = delta_h[k] # set delta h from 0 to 5
#         h = samples_MC-1.5-del_h #adjust height of water based on height of␣
 ↪home above river (1.5) and height of house level adjustment (delta_h)

#         # calculate damages in dollars, when water exceeds threshold:
#         damages = []
#         for i in range(0,len(h)):
#             if h[i] > 0:
#                 damage = (1/(1+np.exp(-k*(h[i]-x_0))))*400000 #convert to␣
 ↪dollars
#                 damages.append(damage)
#         damages = np.array(damages)
#         #Now for each year in 30 years - calculate damages and associated NPV:
#         NPV = 0
#         for j in np.arange(0,T):
#             t = j
#             NPV = NPV + (damages[j] * (1- gamma)^(j+1))
#             # after looping through this loop 30 times, reulsting NPV is NPV␣
 ↪at end of 30 years

#         # Store resulting NPV to dataframe as a value under the column␣
 ↪associated with that delta h
```

[38]:
```
# Parameters
random.seed(44)
T = 30 # number of years per simulation
gamma = 0.04 #discount rate
k = 1.25
x_0 = 2
delta_h_values = np.linspace(0, 5, 11) #housing elevation options to consider
num_simulations = 10000

# Placeholder for results
results = []

# MC simulations
for sim in range(num_simulations):
    samples_MC = stats.lognorm.rvs(s = shape, loc=loc, scale=scale, size=T)

    for del_h in delta_h_values:
        # Adjust height of water relative to house
        h = samples_MC - 1.5 - del_h
```

```
        # Compute damages - when h exceeds threshold governed by relationship␣
↪above, damages are incurred
        damages_MC = np.where(h > 0, (1 / (1 + np.exp(-k * (h - x_0)))) *␣
↪400000, 0)

        # Compute NPV (using matrix multiplication - way more efficient haha)
        NPV = np.sum(damages_MC * (1 - gamma) ** np.arange(1, T + 1))
        if del_h == 0:
            baseline_damages = NPV

        cost = 0
        if del_h> 0:
            cost = 100_000 + 2_000*del_h

        damage_compared_baseline = baseline_damages - NPV

        net = damage_compared_baseline-cost

        # Store result in df
        results.append({"sim": sim, "delta_h": del_h, "NPV": NPV,␣
↪"damage_compared_baseline":damage_compared_baseline, "cost":cost, "net":␣
↪net})

# Convert to DataFrame
results_df = pd.DataFrame(results)

# Display sample
# print(results_df.head(15))
```

[39]:
```
plt.hist(results_df[results_df['delta_h']==0]['NPV'], label = '0', alpha = 0.8)
plt.hist(results_df[results_df['delta_h']==1]['NPV'], label = '1', alpha = 0.8)
plt.hist(results_df[results_df['delta_h']==2]['NPV'], label = '2', alpha = 0.8)
plt.hist(results_df[results_df['delta_h']==3]['NPV'], label = '3', alpha = 0.8)
plt.hist(results_df[results_df['delta_h']==4]['NPV'], label = '4', alpha = 0.8)
plt.hist(results_df[results_df['delta_h']==5]['NPV'], label = '5', alpha = 0.8)
plt.legend()
plt.xlabel('Net Present Value (NPV) of Damages ($)')
plt.ylabel(f'Frequency out of {num_simulations} 30-year Simulations')
plt.title(f'NPV for 30-year simulations with varying house height adjustments')
```

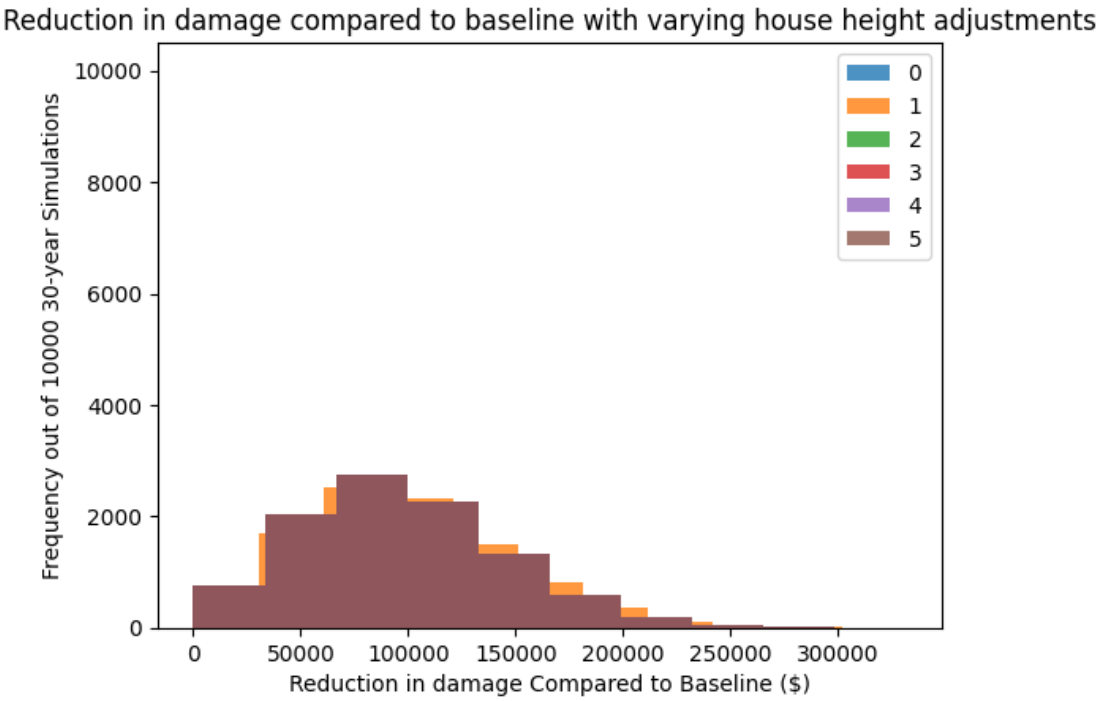
[39]:
```
Text(0.5, 1.0, 'NPV for 30-year simulations with varying house height
adjustments')
```



^This figues shows that damages are mostly accrued for zero height elevation scenario, larger height elevations result in much smaller NPV of damages.

[40]:
```
plt.hist(results_df[results_df['delta_h']==0]['damage_compared_baseline'],␣
 ↪label = '0', alpha = 0.8)
plt.hist(results_df[results_df['delta_h']==1]['damage_compared_baseline'],␣
 ↪label = '1', alpha = 0.8)
plt.hist(results_df[results_df['delta_h']==2]['damage_compared_baseline'],␣
 ↪label = '2', alpha = 0.8)
plt.hist(results_df[results_df['delta_h']==3]['damage_compared_baseline'],␣
 ↪label = '3', alpha = 0.8)
plt.hist(results_df[results_df['delta_h']==4]['damage_compared_baseline'],␣
 ↪label = '4', alpha = 0.8)
plt.hist(results_df[results_df['delta_h']==5]['damage_compared_baseline'],␣
 ↪label = '5', alpha = 0.8)
plt.legend()
plt.xlabel('Reduction in damage Compared to Baseline ($)')
plt.ylabel(f'Frequency out of {num_simulations} 30-year Simulations')
plt.title(f'Reduction in damage compared to baseline with varying house height␣
 ↪adjustments')
```

[40]:
```
Text(0.5, 1.0, 'Reduction in damage compared to baseline with varying house
height adjustments')
```

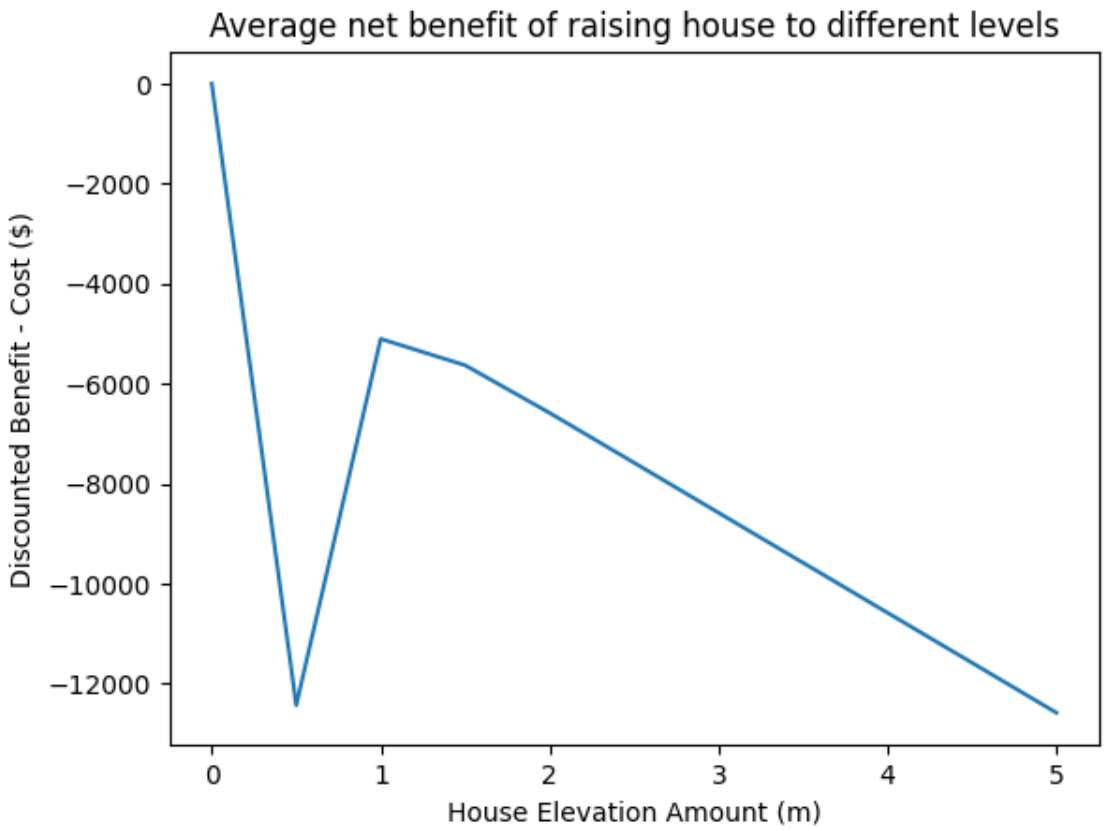


^ This figures shows that elevating the house can dramatically reduce damage (basically in a similar way for all elevation scenarios, as seen by largely overlappin ghistograms).

[41]:
```
average_values = results_df.groupby('delta_h', as_index=False)['net'].mean()
# average_values
```

[42]:
```
plt.plot(average_values['delta_h'], average_values['net'])
plt.ylabel('Discounted Benefit - Cost ($)')
plt.xlabel('House Elevation Amount (m)')
plt.title(f'Average net benefit of raising house to different levels')
```

[42]:
```
Text(0.5, 1.0, 'Average net benefit of raising house to different levels')
```



^The figure above shows that the net difference between the amount "saved" in damages (i.e., baseline scenario NPV damages minus a given elevated scenario NPV damages) is not larger than the cost of elevation for any of the elevation scenarios considered. (i.e., the cost always outweighs the benefit).

[43]:
```
# calculating and plotting expected damage over simulations (e.g., for 0 height␣
↪difference) to see MC convergence:
sim_averages = results_df.groupby(['delta_h', 'sim'], as_index=False)[['NPV',␣
↪'net']].mean() # and also running mean and standard deviation of each of␣
↪these, e.g., NPV for a specific delta_h running average over simulations
sim_averages
```

[43]:

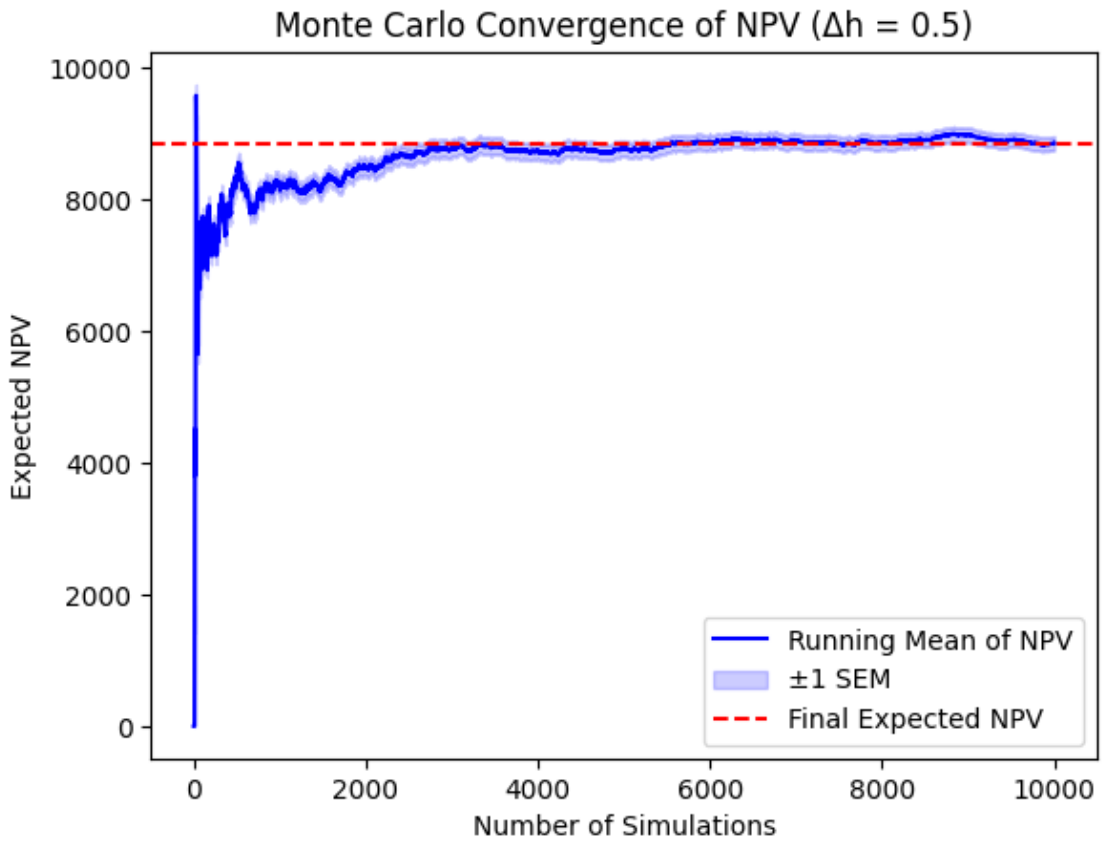
	delta_h	sim	NPV	net
0	0.0	0	81732.672874	0.000000
1	0.0	1	0.000000	0.000000
2	0.0	2	134895.722129	0.000000
3	0.0	3	118165.216897	0.000000
4	0.0	4	129732.978169	0.000000
…	…	…	…	…
109995	5.0	9995	0.000000	-54474.407204
109996	5.0	9996	0.000000	-64446.989049

```
109997      5.0  9997       0.000000 -64417.312934
109998      5.0  9998       0.000000   6192.980503
109999      5.0  9999       0.000000  47342.254334

[110000 rows x 4 columns]
```

USED CHATGPT (OpenAI, 2025): For the code below - I was having trouble finding the right way to calculate a cumulative average and standard error for the mean and asked chatgpt to help, so that I could look at MC convergence (which we're not asked to do in the question but I was just curious about and didn't want to spend too long working on the code for).

[44]:
```
df_sub = sim_averages[sim_averages['delta_h'] == 0.5].sort_values('sim')

# Compute running mean and standard deviation of NPV over simulations
df_sub['running_mean_NPV'] = df_sub['NPV'].expanding().mean()
df_sub['running_std_NPV'] = df_sub['NPV'].expanding().std()

# Compute running standard error (SEM)
df_sub['running_sem_NPV'] = df_sub['running_std_NPV'] / np.sqrt(df_sub.index +␣
 ↪1)  # sqrt of sample size
```

[45]:
```
# Plot running mean
plt.plot(df_sub['sim'], df_sub['running_mean_NPV'], label='Running Mean of␣
 ↪NPV', color='blue')

# Plot confidence intervals (±1 SEM)
plt.fill_between(df_sub['sim'],
                 df_sub['running_mean_NPV'] - df_sub['running_sem_NPV'],
                 df_sub['running_mean_NPV'] + df_sub['running_sem_NPV'],
                 color='blue', alpha=0.2, label='±1 SEM')

# Horizontal line for final expected NPV
plt.axhline(y=df_sub['running_mean_NPV'].iloc[-1], color='red', linestyle='--',␣
 ↪label='Final Expected NPV')

# Labels and title
plt.xlabel("Number of Simulations")
plt.ylabel("Expected NPV")
plt.legend()
plt.title(f"Monte Carlo Convergence of NPV (Δh = 0.5)")
plt.show()
```



^ Here's an example of just one of the NPV damage assessments, for a change in height of half a meter. It shows that the NPV damages converges to somewhere around $9,000; since we know the cost of raising the house by half a meter is much higher than this, it is not surprising that the net benefit-cost is negative.

COMMENTARY - RE: What elevation heights pass the cost-benefit test? What height maximizes the NPV of net benefits (benefits - costs)?

Assuming all costs of elevation are up front (i.e., in year zero, so no discounting applied), the cost benefit analysis was conducted such that the "benefit" is the difference in damages for the baseline scenario vs. a given house elevation scenario (i.e., the amount you "save" in damages if your house is e.g., 1 meter higher vs. if you don't raise it at all), and the cost is the cost of that elevation. The monte carlo simulation above was used to estimate the NPV of the benefits of elevation levels between 0 and 5m. **Based on the main figure above (the one titled 'Average net benefit of raising house to different levels'), none of the elevation heights pass the cost-benefit test.** (Based on the ed discussion post, this is very sensitive to the exact values found for the parmeters of the log normal distirbution, slightly different parameters would have resulted in slightly more frequent "extremes" which would have resulted in at least some of the heights having positive benefit-cost values, however with these parameters, all beenfit-cost are negative besides zero which is simply zero). **An elevation of 0m maximizes the NPV of net benefits (benefits - costs), since all other hieght adjustments have a negative benefit - cost,**

(i.e., the cost outweighs the benefit in each of these other scenarios).

1.4 References

OpenAI. (2025). ChatGPT (March 12 version) [Large language model]. https://chatgpt.com/